Deliverable 1 – Proposal Document

Understanding Revenue Drivers and Customer Purchasing Behaviour in an Online Retail Business

Course: AltSchool Data Science – Second Semester Project

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1. Introduction

Online retail businesses generate large volumes of transactional data through daily customer purchases. While overall sales totals may be available, these figures alone do not provide sufficient insight into what drives revenue, how purchasing behaviour changes over time, or whether certain customer groups contribute more significantly to business performance.

1.1 Background of the Study

The rapid growth of e-commerce has transformed how businesses interact with customers and manage sales operations. Organizations now rely heavily on data generated from digital transactions to understand customer preferences, optimize pricing strategies, and improve overall business performance. However, having large volumes of data does not automatically translate into actionable insight.

2. Problem Statement

Despite having access to large amounts of transactional data, many online retail businesses struggle to extract meaningful insights that inform strategic decisions. Management lacks clear answers regarding revenue drivers, revenue trends, and customer value.

3. Analytical Objectives

1. Identify factors associated with higher revenue.
2. Analyze revenue trends over time.
3. Compare revenue contributions between repeat and one-time customers.
4. Translate analytical findings into actionable business insights.

4. Data Source Description

The dataset used for this analysis is the Online Retail dataset obtained from the UCI Machine Learning Repository (archive.ics.uci.edu). The dataset contains transactional records for a UK-based online retail company.

1. Subject Area: Business / E-commerce
2. Dataset Type: Multivariate, Sequential, Time-Series
3. Time Period Covered: 01/12/2010 – 09/12/2011
4. Number of Instances: 541,909 transactions
5. Feature Types: Integer and Real-valued variables
6. Associated Analytical Tasks: Classification and Clustering

5. Key Variables of Interest

Revenue, InvoiceDate, Quantity, UnitPrice, and CustomerID.

6. Proposed Metrics

1. Total revenue
2. Monthly revenue
3. Average revenue per customer
4. Revenue contribution by customer segment

7. Proposed Statistical Methods

Descriptive statistics, time-series aggregation, outlier detection using IQR, group comparisons, and basic inferential statistics.

8. Assumptions

The dataset accurately represents customer transactions and calculated revenue reflects actual sales value.

9. Limitations

The dataset covers a limited time period and does not include customer demographic information.

10. Expected Outcomes

The project is expected to identify major revenue drivers, reveal trends in revenue over time, and demonstrate whether repeat customers contribute more revenue than one-time customers.